

A practical guide to Bayesian group sequential designs[†]

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Bayesian approaches to the monitoring of group sequential designs have two main advantages compared with classical group sequential designs: first, they facilitate implementation of interim success and futility criteria that are tailored to the subsequent decision making, and second, they allow inclusion of prior information on the treatment difference and on the control group. A general class of Bayesian group sequential designs is presented, where multiple criteria based on the posterior distribution can be defined to reflect clinically meaningful decision criteria on whether to stop or continue the trial at the interim analyses. To evaluate the frequentist operating characteristics of these designs, both simulation methods and numerical integration methods are proposed, as implemented in the corresponding R package *gsbDesign*. Normal approximations are used to allow fast calculation of these characteristics for various endpoints. The practical implementation of the approach is illustrated with several clinical trial examples from different phases of drug development, with various endpoints, and informative priors. Copyright © 2013 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Classical group sequential designs are well established [1–3] and widely used in confirmatory clinical trials [4–6]. In these adaptive designs, patients are typically randomized to an experimental treatment or placebo, and at the interim analyses, decisions are taken to either stop the trial for success/futility or to continue. In these classical group sequential designs, a clinical trial is considered a success, if the experimental treatment is statistically significantly better than placebo. The criteria for stopping or continuing the trial are chosen to control the false positive rate (type I error). Criteria that are directly related to the observed treatment effect at the end of the confirmatory trial are currently not considered in the literature on classical group sequential design methodology. However, the observed treatment effect is a key element for benefit–risk assessment by health authorities, for evaluating whether the drug should be reimbursed, and for market success.

In Bayesian group sequential designs [7, 8], success and futility stopping criteria that are related to both treatment effect size and superior efficacy against placebo are commonly used. Criteria are often based on the posterior distribution of the treatment effect at the interim analyses, although predictive criteria have also been considered. Bayesian group sequential designs are particularly well suited to clinical studies in earlier phases of drug development, for example, in proof-of-concept (PoC) trials or in phase II oncology trials. These studies are designed to enable company internal decisions on whether to further invest in the development of the experimental drug. Decisions in this context are typically centered on the size of the treatment effect, and hence, it is important that the stop/go criteria used in the trial design reflect this. An advantage of the Bayesian approach is that prior information can be included in the analysis. For example,

if similar previous placebo-controlled studies are available, then this historical placebo information can be included as appropriately down-weighted prior information and hence may allow us to reduce the number of placebo patients in the planned trial.

Bayesian group sequential methods are not designed to optimize frequentist operating characteristics but rather to provide stopping criteria related to the subsequent decision making. However, it is generally recognized that evaluation of frequentist operating characteristics is also relevant for Bayesian clinical trial designs [8]. The main characteristics of interest are the probability to stop for success, to stop for futility, or to continue for each interim analysis and the final analysis, as well as the corresponding cumulative probabilities. The expected sample size is another characteristic useful to quantify the gain of the group sequential design as compared with a fixed design. These operating characteristics can be evaluated either by simulation or by numerical integration, for scenarios corresponding to different true values of the treatment effects.

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In Section 2, we describe a general class of Bayesian group sequential designs and the evaluation of their operating characteristics based on normal approximations. Stopping criteria and the use of prior information are also discussed. In Section 3, several examples illustrate the use of this approach in the design of clinical trials, from different phases of drug development, and various endpoints. The R package *gsbDesign* (group sequential Bayesian design) [9] is used for the evaluation of operating characteristics, and the corresponding R code is provided as supplementary information. The paper closes with a discussion.

2. METHODS

2.1. Effect and probability thresholds

Clinical trials are used for decision making by investigators and health authorities, and the observed treatment effect is highly relevant in this context. However, clinical trials are often designed as if statistical significance were the sole measure of success. That is, traditionally null ($H_0 : \delta \leq 0$) and alternative ($H_1 : \delta > 0$) hypotheses are formulated for the true treatment difference δ , and the type I error (α) is set to a specified value (large values of δ correspond to a positive effect). To power the study at a specified level $1 - \beta$, investigators decide on a treatment effect size (δ_1), often interpreted as a minimal clinically relevant effect. The null hypothesis will be rejected if the observed one-sided p -value is less than α . On the basis of planning assumptions, it is possible to re-express this criterion in terms of the observed average treatment effect required to reject the null. For instance, in the standard two-group normal case with known standard deviation, the null hypothesis will be rejected if the observed average treatment effect exceeds the threshold $D_T = \delta_1 z_{1-\alpha} / (z_{1-\alpha} + z_{1-\beta})$, where z_ε is the 100ε -th quantile of the standard normal distribution. This threshold D_T is typically smaller than δ_1 ; for example, for $\alpha = 0.05$ and $\beta = 0.2$, D_T is $0.67 \delta_1$. Thus, the null hypothesis can be rejected for observed treatment effects that are much smaller than δ_1 , and hence for effect sizes that are not seen as clinically relevant.

From both frequentist and Bayesian perspectives, statistical significance is arguably not sufficient as a measure of success for clinical trials [10–16]. Within a frequentist framework, the additional requirement of a minimal observed difference δ^* has often been proposed. This double criterion (significance and minimal observed effect size) encourages trial designs that avoid situations where a statistically significant result is obtained with a small observed treatment effect.

From a Bayesian perspective, the decision making is based on the posterior probability for the treatment effect given the clinical trial data. We consider here success criteria for Bayesian designs of the form:

$$p(\delta > s_1 | \text{Data}) > p_1 \quad \text{and} \quad p(\delta > s_2 | \text{Data}) > p_2,$$

where s_1 and s_2 are specified effect thresholds and p_1 and p_2 are specified probability thresholds. Some flexibility in the choice of the effect and probability thresholds is possible, as the criteria relate to two particular quantiles of the posterior distribution only. For example, with normally distributed posteriors, the posterior with minimal precision satisfying the double criteria has expectation and standard deviation given by

$$E(\delta | \text{Data}) = (s_1 z_{1-p_2} - s_2 z_{1-p_1}) / (z_{1-p_2} - z_{1-p_1}),$$

$$Sd(\delta | \text{Data}) = (s_2 - s_1) / (z_{1-p_2} - z_{1-p_1}).$$

As any two quantiles completely define a normal distribution, this allows us to express the double criterion using other quantiles of interest. Thus, for ease of interpretation and communication with clinicians, it is often helpful to choose zero as the first effect threshold and 0.5 as the second probability threshold, so that the success criterion is

$$p(\delta > 0 | \text{Data}) > 1 - \alpha \quad \text{and} \quad p(\delta > \delta^* | \text{Data}) > 0.5.$$

It should be noted that this then approximately corresponds to the frequentist double criterion (significance and minimal observed effect size), when only vague prior information is available. Hence, the Bayesian design becomes particularly relevant when prior information is available, either for the treatment effect δ or for both treatment arms individually [17].

It is often useful to also define criteria for futility. We consider Bayesian futility criteria of the form

$$p(\delta < f_1 | \text{Data}) > q_1 \quad \text{and} \quad p(\delta < f_2 | \text{Data}) > q_2,$$

where f_1 and f_2 are specified effect thresholds and q_1 and q_2 are specified probability thresholds. These criteria are particularly useful if the clinical trial contains interim analyses, where early stopping for futility is attractive from an ethical and financial perspective. However, futility criteria can also be considered in the final analysis, to classify a trial as a clear success, a clear failure, or an indeterminate outcome, that is, when none of the success and none of the futility criteria are fulfilled. An indeterminate situation is undesirable as one may then need to collect further information, for example, from additional clinical trial(s).

2.2. Interim analyses and operating characteristics

In the following, we consider two-arm clinical trials with one or more interim analyses, and at each interim analysis, success and futility criteria are evaluated to decide if the trial should be stopped. The criteria are based on the posterior distribution of the treatment effect δ , as described in Section 2.1. For illustration, suppose that a clinical trial with three interim analyses and a final analysis is planned, with no stopping for futility, aggressive stopping for success, and no relevant prior information. Then at each interim analysis, success may be declared if both $p(\delta > 0 | \text{Data}) > 1 - 0.009$ and $p(\delta > 4 | \text{Data}) > 0.5$, say. Here the first criterion corresponds to a classical Pocock stopping rule (statistical significance testing), and the second criterion essentially requires that the observed effect size is at least four units. The double success criterion can be translated to stopping boundaries on the observed difference between the treatments. Figure 1 shows these boundaries, which can be constructed as the maximum of the two stopping boundaries corresponding to the two criteria related to significance testing and effect size, respectively. It can be seen here that the first criterion related to statistical significance dominates in the first phase, whereas the second criterion is the dominating criterion in the second phase of the trial. For example, at the first interim analysis, the first criterion is a higher hurdle for the treatment difference, and whenever the first criterion is met, then the criterion related to effect size will be met as well. At the final analysis, achieving the second criterion on the required effect size is a higher hurdle than the significance test. Less aggressive stopping rules can be obtained if higher hurdles for success are used in the earlier interim analyses.

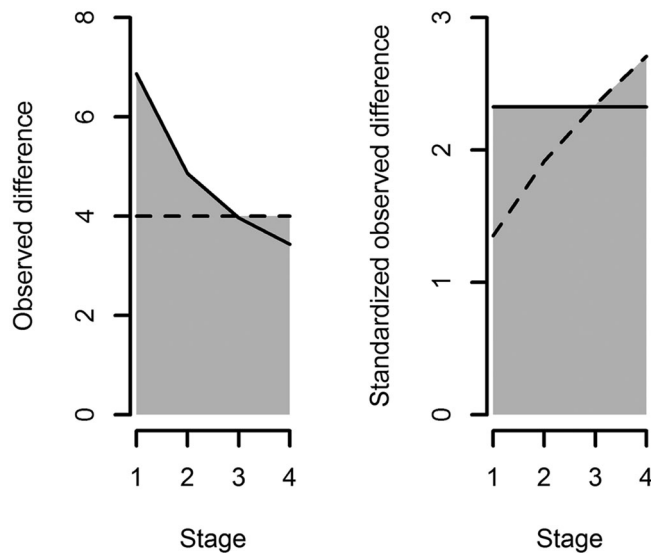


Figure 1. Stopping boundaries for a clinical trial with three interim analyses and a final analysis, using a dual success criterion. The first criterion is related to statistical significance (solid line) and the second one to observed effect size (dashed line). The trial is a success if both criteria hold (observed difference above gray area).

For the interim and final analyses of the clinical trial, Bayesian evaluations of the posterior distribution may require use of Markov chain Monte Carlo methods. Although this is not a problem for the analysis of the actual trial, calculation of frequentist properties through simulation would then often take considerable time. For the evaluation of operating characteristics, we therefore use approximate normalized likelihoods [7] and normal priors, as illustrated in Section 3 for binary, count, and time-to-event endpoints. The posterior distributions can then be evaluated analytically. More precisely, let $Y_{kij} \sim N(\theta_k, \sigma^2)$ denote the observations for treatment $k=1,2$ in stage $i=1, \dots, I$, for subject $j=1, \dots, n_{ki}$, with known standard deviation σ . The treatment effect is then $\delta = \theta_2 - \theta_1$, and prior information may be available on the difference δ , or on the two arms (θ_1 and/or θ_2). If prior information on the treatment effect is given as $\delta \sim N(\delta_0, \sigma_0^2)$, then the posterior distribution at stage i is given by

$$\delta | \text{stage } i \sim N\left(\omega_i \delta_0 + (1 - \omega_i) D_i, 1 / \left\{1 / \sigma_0^2 + 1 / V_i\right\}\right),$$

where D_i is the aggregated treatment effect at stage i , V_i is the corresponding variance, and $\omega_i = 1 / (\sigma_0^2 / V_i + 1)$. If prior information on the two arms is available as $\theta_1 \sim N(\theta_{10}, \sigma_1^2)$, and $\theta_2 \sim N(\theta_{20}, \sigma_2^2)$, then the posterior distribution of the treatment effect δ at stage i is given by

$$\begin{aligned} \delta | \text{stage } i \sim N\left(\left\{\omega_{2i} \theta_{20} + (1 - \omega_{2i}) Y_{2i}\right\} \right. \\ \left. - \left\{\omega_{1i} \theta_{10} + (1 - \omega_{1i}) Y_{1i}\right\}, 1 / \left\{1 / \sigma_1^2 + 1 / V_{1i}\right\} \right. \\ \left. + 1 / \left\{1 / \sigma_2^2 + 1 / V_{2i}\right\}\right), \end{aligned}$$

where Y_{ki} is the aggregated mean in arm k at stage i , V_{ki} is the corresponding variance, and $\omega_{ki} = 1 / (\sigma_k^2 / V_{ki} + 1)$.

For such Bayesian group sequential designs, simulation methods can be used to evaluate the operating characteristics. For given true effect parameters μ_1 and μ_2 , normally distributed data Y_{kij} are generated, the posterior distribution at each stage is derived, and it is checked whether the success or futility

criteria are fulfilled. Repeating this process many times, the operating characteristics can be calculated with the desired precision.

In Section 3, the evaluation of operating characteristics is illustrated for several examples of clinical trials, making use of the R package *gsbDesign*, which can be downloaded at CRAN (cran.r-project.org). In this package, faster numerical integration rather than simulation is used if feasible. For a detailed description of the technical details, see Gerber et al. [9].

2.3. Prior information

2.3.1. Using historical information on the control. When planning a clinical trial comparing an experimental treatment with a control, historical information on the effect of the control treatment may be available. Several methods have been proposed to derive from this a prior distribution for the control effect [18–20]. We consider here the meta-analytic-predictive approach [21], which can be used when historical randomized trials with a similar design comparing other experimental treatments with the control are available. On the basis of a random-effects model for the true control effects in the historical and the planned study, the predictive distribution of the true control effect in the planned study is then used as the prior distribution. The random-effects model assures that the historical information is down-weighted appropriately, depending on the between-trial variability. The approach is illustrated in Section 3.1.

2.3.2. Priors on the treatment effect. Historical information on the difference between experimental treatment and control is sometimes available. For example, several randomized clinical trials directly or indirectly linked to the experimental or control treatment may be available, and then a network meta-analytic-predictive approach [22] can be applied to derive an appropriate prior distribution for the difference of the two treatments, taking into account between-trial variability.

Rather than using priors for the treatment difference derived from historical information, archetypal priors reflecting a skeptical or an optimistic person may be useful [7]. For example, a clinical trial may only be stopped for futility, if even using an optimistic prior, the trial seems not worth pursuing; an illustrative case is presented in Section 3.3.

3. EXAMPLES

In the following, a series of examples illustrate the broad utility of the approach, in various phases of drug development, with normal, binary, count, and time-to-event endpoints. The operating characteristics of the designs were evaluated using the R package *gsbDesign* [9], and the code for all the examples is provided as supplementary information (refer to the Supporting Information).

3.1. Design of a proof-of-concept trial in Crohn's disease

A PoC study is the first clinical trial that provides evidence on the efficacy of a new experimental treatment. On the basis of the results of the PoC study, the sponsor decides on whether to stop or continue further development. We consider the design of a PoC study in Crohn's disease [23], an inflammatory bowel disease, where the primary endpoint is the change from baseline to week 6 in the Crohn's Disease Activity Index (a decrease corresponds to an improvement). For the planning of the trial, the standard deviation of this primary endpoint is assumed to be $\sigma = 88$, based on

previous studies. The PoC study should be designed such that it allows informed decision making. First, it should provide clear evidence that the experimental drug is better than placebo. Second, the effect should be sufficiently large to warrant further development. After discussions within the clinical team, and taking account of effects seen with marketed drugs in Crohn's disease, the success criteria were quantified as

$$p(\delta > 0 | \text{Data}) \geq 0.95 \quad \text{and} \quad p(\delta > 50 | \text{Data}) \geq 0.50, \quad (1)$$

where $\delta = \theta_P - \theta_E$ denotes the true difference between placebo and experimental treatment on the primary endpoint. In a frequentist framework, the first of these criteria approximately corresponds to a one-sided significance test ($\alpha = 0.05$), and the second criterion approximately corresponds to the requirement that the observed difference is at least 50. No reliable prior information on the effect of the experimental treatment was available, and hence, a non-informative prior was chosen for θ_E , that is, $\theta_E \sim N(0, \sigma^2/0.001)$. For placebo treatment, an informative prior was derived from six placebo-controlled historical studies in Crohn's disease, using a meta-analytical-predictive approach (Section 2.3). The main assumption here is that the placebo effects in the six historical studies ($\theta_{P1}, \dots, \theta_{P6}$) and in the PoC study (θ_P) are exchangeable, as expressed by a random-effects model:

$$\theta_P, \theta_{P1}, \dots, \theta_{P6} \sim N(\mu, \tau^2).$$

The between-trial standard deviation τ quantifies the degree of similarity of the true placebo effect in the various studies. A random-effects meta-analysis of the six historical studies provides a posterior predictive distribution for the true placebo effect θ_P in the new study (Figure 2). A normal approximation of this posterior predictive distribution is then used as the prior for θ_P , that is, $\theta_P \sim N(-49, \sigma^2/20)$. This prior information is equivalent to about 20 virtual patients, with an average Crohn's Disease Activity Index decrease from baseline of 49. This corresponds to a considerable down-weighting of the 671 historical placebo patients due to between-trial variability τ , whose posterior median (95%-interval) is 13 (2,40). The informative prior on the placebo effect allows us to reduce the number of placebo patients in the PoC study.

A two-arm Bayesian group sequential design with one interim analysis was considered appropriate for the PoC study, comparing the experimental treatment with placebo. To allow for early stopping for futility, a futility criterion was defined as

$$p(\delta < 40 | \text{Data}) \geq 0.90. \quad (2)$$

Evaluations of the operating characteristics for various sample sizes and different timings of the interim analysis finally led to the

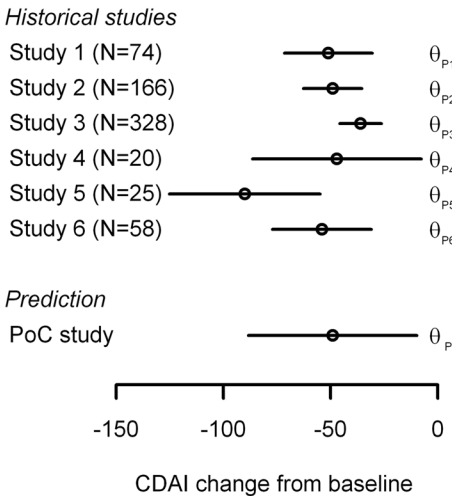


Figure 2. Random-effects meta-analysis of the placebo effect in six historical studies in Crohn's disease, and the predicted true placebo effect in the proof-of-concept study.

following two-stage design. In the first stage, 10 patients are allocated to placebo and 20 patients to the experimental treatment. If the success criteria (1) are both satisfied, the PoC study will stop for success. If the futility criterion (2) is satisfied, the study will stop for futility. If neither (1) nor (2) occurs, then another 10 patients are allocated to placebo and another 20 patients to the experimental treatment. At the final analysis, again both success and futility criteria are evaluated, allowing for possibly indeterminate outcomes, that is, when neither successful nor futile results are obtained. The operating characteristics of this design were evaluated using gsbDesign [9] and are summarized in Table I (for a true placebo effect of $\theta_P = -49$). If the experimental treatment is placebo-like, then the PoC will be declared successful in only 1.2% of cases; that is, the type I error is low. If the experimental treatment is borderline effective ($\delta = 50$) or similar to competitors ($\delta = 60$), then a successful PoC is expected in 62.7% and 80.8% of cases, respectively. It should be noted that such aggressive stopping rules are typical for early phase trials. The design provides considerable savings in the number of patients, compared with a fixed design using no prior information ($n = 80$). The expected sample size is typically between 35 and 50, depending on the true effect size.

3.2. Oncology phase II single-arm trials

The Simon two-stage design [24] is routinely used for single-arm oncology phase II trials where the objective is to decide whether

Table I. Operating characteristics of the two-stage design for the proof-of-concept trial in Crohn's disease.										
Difference δ	Probability (%) interim			Probability (%) final			Probability (%) overall			Expected <i>N</i>
	Success	Indeterminate	Futile	Success	Indeterminate	Futile	Success	Indeterminate	Futile	
0	1.1	35.6	63.3	0.1	78.6	21.3	1.2	14.1	84.7	40.7
40	32.2	61.0	6.8	8.9	87.1	4.0	41.1	48.1	10.8	48.3
50	50.0	47.5	2.5	12.7	86.1	1.2	62.7	33.5	3.8	44.2
60	67.6	31.6	0.8	13.3	86.4	0.3	80.8	18.1	1.1	39.5
70	82.2	17.6	0.2	10.3	89.7	0.0	92.5	7.3	0.2	35.3

Table II. Examples of Simon's two-stage design, with expected sample size and probability of early termination for a true response rate of the experimental treatment $\pi_E = \pi_C^*$ ($\alpha = 0.05, \beta = 0.20, \pi_C^* = 0.4, \pi_E^* = 0.6$).

Design	Rejection criteria $\leq r_{E1}/n_{E1}$ $\leq (r_{E1} + r_{E2})/(n_{E1} + n_{E2})$	Expected sample size	Probability of early termination
Optimal	7/16	24.5	0.72
Minimax	17/34	34.4	0.91

Here r_{E1} and r_{E2} (n_{E1} and n_{E2}) are the numbers of responders (patients) in the first and second stages, respectively.

a new experimental treatment has sufficient activity to invest in further development. The two-stage design allows early stopping for futility but no early stopping for success. To define the design, one has to specify type I error (α), type II error (β), and two relevant response rates π_C^* and π_E^* . Simon's two-stage design is based on a statistical significance test. If the true underlying response rate of the new experimental treatment is less than π_C^* (assumed historical control rate), it should be unlikely (low type I error) to judge the new drug sufficiently promising for further development. On the other hand, if the true underlying response rate of the new treatment is larger than π_E^* , then it should be likely (high power) to judge the new drug sufficiently promising. The design aims at minimizing the number of patients treated with a drug with low activity. Simon [24] suggested two optimal designs: one minimizing the expected sample size (optimal design) and one minimizing the maximum sample size (minimax design); examples of such designs are shown in Table II.

The paper by Simon [24] provided only limited information on the operating characteristics of the two-stage design. More complete operating characteristics can be obtained with gsbDesign [9], by reformulating the single-arm design as a two-arm design, and using a normal approximation for the endpoint. For binary data, the normal approximation is known to be more appropriate on the logit scale than on the proportion scale. Let r_{Ei} be the number of responses and n_{Ei} the number of treated patients in stage i , $i = 1, 2$ of the experimental arm. Defining y_{Ei} as the corresponding log-odds of the observed response rate, its distribution is approximately given by

$$y_{Ei} \sim N(\text{logit}(\pi_E), \sigma_E^2/n_{Ei}),$$

where π_E is the true response rate and $\sigma_E^2 = 1/\pi_E + 1/(1 - \pi_E)$. It should be noted that for small response rate π_E and small number of patients, this normal approximation may not be appropriate.

To mimic Simon's two-stage design, we introduce a virtual control arm, into which no patients are recruited, that is, $n_{C1} = n_{C2} = 0$, but for which an extremely informative prior for the response rate π_C is used, that is, a prior centered at π_C^* with an almost zero variance.

For example, let $\pi_C^* = 0.4, \pi_E^* = 0.6, \alpha = 0.05$, and $\beta = 0.20$. According to Simon's optimal two stage design (Table II), $n_{E1} = 16$ patients are recruited in the first stage and $n_{E2} = 30$ in the second stage. The trial is stopped for futility at the interim analysis if seven or less patients respond. The trial is a success at the end of the trial if more than 23 patients respond. Translating these criteria to the log-odds scale, the futility criterion is

$$p(\delta < \text{logit}(7.5/16) - \text{logit}(0.4)|\text{Data}) \geq 0.5, \quad (3)$$

and the success criterion is

$$p(\delta > \text{logit}(23.5/46) - \text{logit}(0.4)|\text{Data}) \geq 0.5, \quad (4)$$

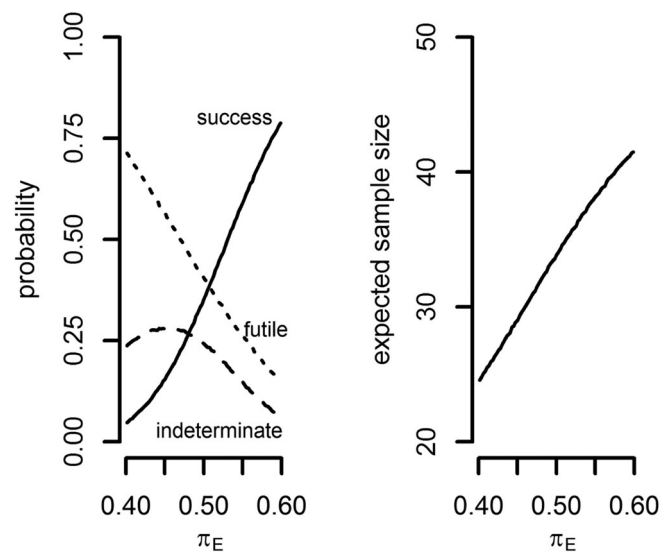


Figure 3. Operating characteristics for a Simon two-stage design ($\alpha = 0.05, \beta = 0.20, \pi_C^* = 0.4, \pi_E^* = 0.6$) for a single-arm phase II oncology trial.

where $\delta = \text{logit}(\pi_E) - \text{logit}(\pi_C)$. Using gsbDesign [9] with these criteria, one obtains an expected sample size of 24.6 and a probability of early termination of 0.71 for $\pi_E = 0.4$; these values are very close to those obtained by the exact calculations performed by Simon [24] (see Table II). Figure 3 shows the corresponding operating characteristics for various values of the true response rate π_E . It should be noted that the operating characteristics in this example are based on normal approximations and hence, will not exactly match those obtained from using the binomial model.

3.3. Design of a phase II trial in multiple sclerosis with count endpoint

Within this section, a phase II study design for a novel treatment for multiple sclerosis (MS) is examined. The Bayesian group sequential approach shall be applied, allowing the possibility to stop early for either futility or success at a single interim analysis. In addition, we consider the impact of an enthusiastic prior, which in practice could be based on information arising from clinical trials with other drugs having the same mechanism of action.

The primary outcomes for phase II MS studies are typically based on magnetic resonance imaging (MRI), usually in the form of lesion counts that are around 5 to 10 times more sensitive than clinical measures for detecting MS activity [25]. Although MRI imaging is not accepted for regulatory approval, it is typically one of the most important factors in drug development decision making. Recent work by Sormani *et al.* [26] showed a strong trial level relationship between the treatment effect expressed as

the relative reduction in MRI lesions (when comparing an experimental treatment with control) and the corresponding treatment effect on the annualized relapse rate, which is typically used as the primary outcome in phase III clinical trials.

When selecting an appropriate sampling model for the analysis of phase II MRI data, Sormani et al. [27] proposed the use of negative binomial (NB) regression. Further in their comparative analysis of various models to study lesion counts, van den Elskamp et al. [28] concluded that the NB distribution is best for modeling new enhancing lesion counts, irrespective of the effect of treatment, follow-up duration, or a baseline activity selection criterion. On the basis of these investigations, a negative binomial model shall be assumed for purposes of planning and trial design in MS.

To implement planning using gsbDesign [9], an approximate normal likelihood must be formed for the estimates of the treatment effect parameters arising from an NB model. Typically, such a model will be fit within a generalized linear modeling framework, using a log link function, where the key parameters of interest will be the log rates associated with each treatment group. This leads to an overall treatment effect on the log relative risk scale that might subsequently be presented as a percentage relative reduction achieved by treatment over control. To develop a normal approximation for the log rate parameters, we utilize the development of Friede and Schmidli [29], who applied the delta method. Specifically, on the basis of the parameterization of a negative binomial distribution given by Keene et al. [30], where κ represents the assumed common over-dispersion parameter and the parameters, μ_j , $j = 1, 2$, denote the underlying true mean MRI counts associated with the treatment ($j = 1$) and control ($j = 2$)

groups, the following normal approximation can be formed for the estimates of the treatment effect parameters, denoted by r_{jj} , $j = 1, 2$, arising from a trial with n_j subjects randomized to treatment group j :

$$\log(r_j) \sim N(\log(\mu_j), 1/(\mu_j n_j) + \kappa/n_j).$$

By noting that it is possible to specify $\sigma_j^2 = 1/\mu_j + \kappa$, this formulation can be used to examine designs within the gsbDesign package [9]. However, some care is required in this situation, as the appropriate patient level variance will vary according to the assumed treatment effect parameters.

Returning to the example, it shall be assumed that a parallel group two-arm phase IIa clinical trial will be conducted, where 100 subjects are to be randomized using a 1:1 ratio, to a novel treatment ($j = 1$) and control ($j = 2$). The primary efficacy variable is based on MRI and will be defined as the cumulative number of new Gadolinium-enhancing T1 lesions from months 1 to 6, which is commonly used in such trials [25]. When conducting the trial, an interim analysis is planned with the possibility to stop early for either futility or success, at a time point where final outcome data are available for 50% of the patients. For the purposes of examining design operating characteristics, we shall fix the over-dispersion parameter κ to be 2 and the average cumulative lesion count associated with placebo to be 14.8 [25].

The formulation of decision rules for futility and success at the interim analysis and for the final analysis, shall be based on the posterior distribution associated with the log relative risk $\delta = \log(\mu_2/\mu_1)$. Specifically, to stop early for success, the following double criterion was deemed to provide reasonable evidence to warrant further development:

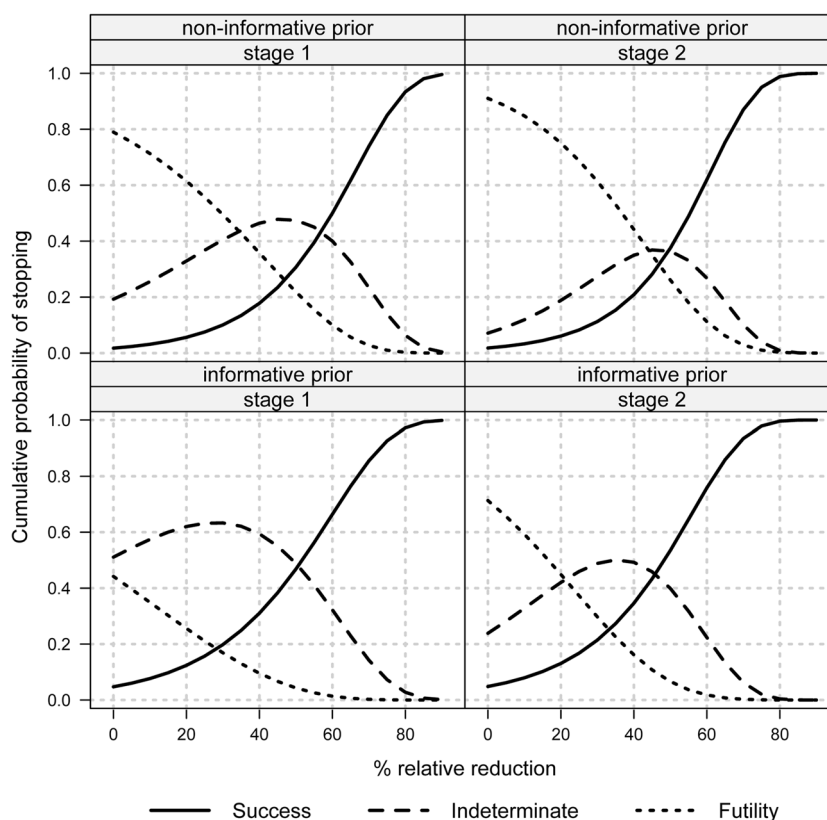


Figure 4. Cumulative probability of stopping for either success or futility for a non-informative prior (upper panel) and an informative prior (lower panel) in a phase II multiple sclerosis trial with count endpoint.

$$p(\delta > 0 | \text{Data}) > 0.9 \quad \text{and} \quad p(\delta > \log(2.5) | \text{Data}) > 0.5.$$

This is equivalent to having a posterior median for δ corresponding to at least a 60% relative reduction, with a 90% posterior probability of some reduction. Whereas, in the case of stopping early for futility, a recommendation to stop would be based on the following double criterion:

$$p(\delta < \log(1.43) | \text{Data}) > 0.5. \quad \text{and} \quad p(\delta < \log(2.5) | \text{Data}) > 0.9.$$

This rule is equivalent to having a posterior median for δ corresponding to a risk reduction of less than 30% and a 90% posterior probability of δ having a risk reduction of less than 60%. Additionally, we assume that prior information is available from a phase II trial conducted in a compound within the same class, where a risk reduction of 75% was observed in a trial consisting of 80 patients per group, with placebo rate and over-dispersion parameter equal to the planning assumptions stated previously. On the basis of this information, two priors were considered for δ in the design evaluation: (1) a non-informative prior, where no prior weight is given to the previous study; and (2) a prior based on discounting the information from the previous study to be worth 10 patients per group.

The resulting operating characteristics for the cumulative probability of success or futility at either the interim analysis (stage 1) or the final analysis (stage 2) are displayed in Figure 4 for a non-informative and informative prior, respectively. The impact of the different priors is evident: use of an informative enthusiastic prior will increase the probability of success and reduce the chance of stopping for futility. This emphasizes that informative prior based on data from another compound within the same drug class must be due to a genuine prior belief of exchangeability of the treatment effects.

3.4. A phase III trial design with Bayesian futility criteria for a time-to-event endpoint

In phase III trials with a time-to-event endpoint, classical group sequential designs are commonly used. The standard approach assumes approximate normal distributions for the underlying test statistics [5], and hence, the *gsbDesign* package [9] can be applied in this context. To illustrate its use, we consider a recent large non-inferiority trial [31]. The study compared a test treatment (aspirin plus extended-release dipyridamole) with an active-control treatment (clopidogrel) in the prevention of recurrent stroke, with the primary endpoint being the recurrence of stroke. The study used a classical group sequential design with two interim analyses and the possibility for stopping using modified Haybittle–Peto bounds for testing the null hypothesis of no difference between the treatment groups. The stopping criterion was based on a p -value of 0.0001 at the first interim analysis and a p -value of 0.001 at the second interim analysis. The trial was hence planned to stop early, if one of the treatments (not necessarily the test treatment) is better than the other. The interim analyses were conducted after one-third and two-thirds of the expected 1800 events had occurred. The final analysis was based on a non-inferiority test between the two treatments with a non-inferiority margin for the ratio of hazard rates of test and control treatment of 1.075, using an unadjusted one-sided 2.5% level of significance. The trial did not stop early. First recurrence of stroke occurred in 916 of 10,181 patients (9%) in the test group and 898 of 10,151 patients (8.8%) in the control group. The hazard ratio under the proportional hazards assumption is then roughly 1.01 with 95% approximate

confidence interval (0.92, 1.11). As the non-inferiority margin was 1.075, non-inferiority could not be shown.

In what follows, we investigate operating characteristics for this trial for an alternative futility rule. For this purpose, we use the standard asymptotic normal approximation for the estimate of the log-hazard ratio. For a $k:1$ randomization, and under the assumption of equal survival distributions in the two groups, the distribution of the estimator of the log-hazard rate is for large sample sizes approximately normally distributed with mean equal to the true log-hazard ratio and variance given by $c/(E_1 + E_2)$, where $c = (k + 1)^2/k$ and E_1 and E_2 are the number of events in test and control group, respectively ([5], ch.3.7; [7], ch.2.4.2). To implement this in the *gsbDesign* package [9], the expected number of events per group and stage corresponds to sample sizes, and the standard deviation in the two groups can be chosen as $\sigma_1 = \sigma_2 = \sqrt{cE_1E_2/(E_1 + E_2)^2}$ to obtain the desired asymptotic distribution. Hence for a 1:1 randomization, $\sigma_1 = \sigma_2 = 1$.

We first formulate the frequentist decision rules as Bayesian decision rules, with an uninformative prior distribution. Denote by λ the hazard ratio. As the p -value was two-sided, the Haybittle–Peto decision rules correspond to stopping for success or futility when

$$\begin{aligned} p(\log(\lambda) < 0 | \text{Data}) &> 1 - 0.0001/2 \\ \text{or} \quad p(\log(\lambda) > 0 | \text{Data}) &> 1 - 0.0001/2 \end{aligned}$$

at the first interim analysis and stopping when

$$\begin{aligned} p(\log(\lambda) < 0 | \text{Data}) &> 1 - 0.001/2 \\ \text{or} \quad p(\log(\lambda) > 0 | \text{Data}) &> 1 - 0.001/2 \end{aligned}$$

at the second interim analysis. The Bayesian version of the final non-inferiority test can be written as

$$p(\log(\lambda) < \log(1.075) | \text{Data}) > 0.975;$$

that is, the non-inferiority margin was 1.075, and the test was performed at one-sided level 0.025.

As alternative futility criteria, we consider rules of the form $p(\log(\lambda) > \log(1.075)) > p$, which allow stopping the trial if one is confident that the treatment effect is worse than the non-inferiority margin. We will investigate the use of $p = 0.95, 0.9, 0.8$ at interim analyses 1, 2 and the final analysis, respectively, and assume that the 1800 total events are equally split between the two groups so that there are 300 events per stage and treatment. As scenarios for the true hazard ratio, we will consider the range 0.9 – 1.15, which is motivated by the actual trial results.

Figure 5 shows the operating characteristics of the design with the original and alternative futility criterion, as evaluated by *gsbDesign* [9]. The alternative futility criterion has only minimal impact on the probability of success (statistical significance) but considerably increases the chance of early stopping when the test treatment is worse than the control, which also affects the expected sample size (in this case calculated as the expected number of events divided by 0.09, which is the expected event rate). For example, if the hazard ratio is 1.15, that is, the test treatment is substantially worse than the control treatment, then more than 3000 patients are saved using the alternative futility rule, which is quite notable keeping in mind that the success curve is largely unaffected by introducing the alternative futility rules.

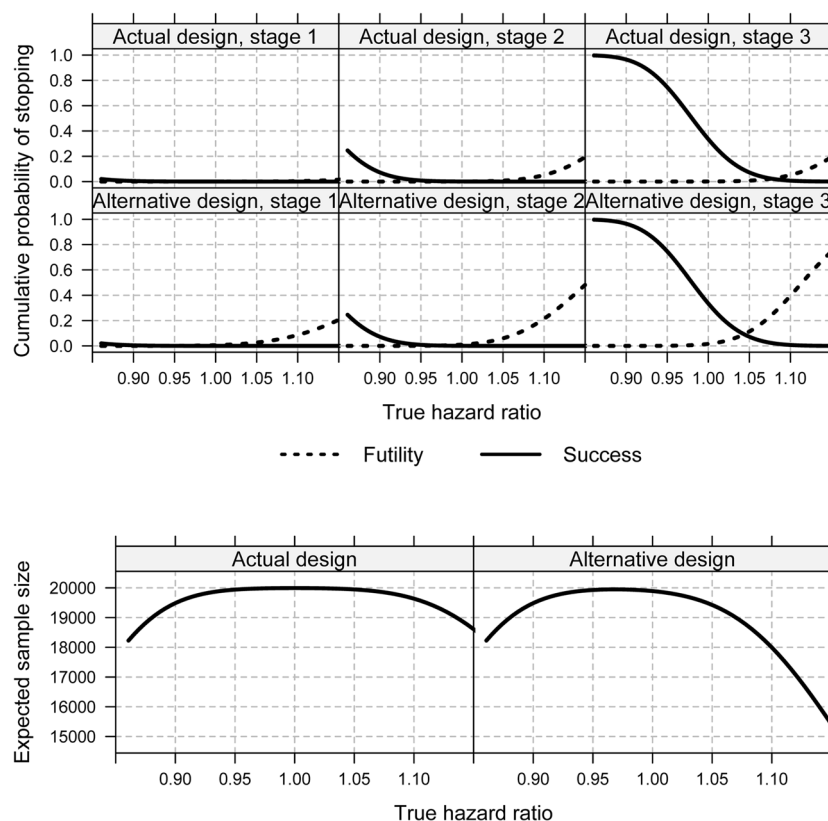


Figure 5. Cumulative probability of stopping and expected sample size for actual and alternative futility rule in a phase III trial with time-to-event endpoint.

In this section, we illustrated the usage of the *gsbDesign* package [9] for the analysis of survival data. Of course there are limitations due to the normal approximation that has been used. For example, if the underlying survival rates are vastly different, specialized code for survival data or direct simulation should be used.

4. DISCUSSION

The Bayesian group sequential approach presented here provides a flexible framework for the design of clinical trials with two treatment arms. Several success and futility criteria can be specified at each interim analysis and hence allow a closer matching of these quantitative stopping rules to clinical decision making. This is particularly useful in early drug development, where clinical trials are mainly used for company-internal decisions. In such settings, criteria are needed that go beyond statistical significance and also relate to the observed effect size. In confirmatory clinical trials, statistical significance is important but not sufficient: additional criteria such as a minimal observed effect size are relevant for late phase trials as well [32].

The Bayesian methodology allows to use prior information on the control group effect or on the difference between experimental and control group effect. Deriving prior information from historical studies requires care, both in the selection of relevant studies and in the (network) meta-analytic methodology. With appropriately chosen studies and a down-weighting of historical information to account for between-trial variability, the risk of conflicts between prior information and the trial data can be minimized. Use of mixture priors can provide robustness against

non-exchangeable treatment effects [7]. Informative priors are particularly useful in smaller early phase trials as the number of patients in the trial may then be considerably reduced [33]. In confirmatory trials, informative priors are the exception. However, as futility stopping does not inflate type I error, Bayesian criteria are often used in this context and then can also include informative priors. The use of archetypal skeptical priors when considering stopping for success has also been suggested for late phase trials [7].

Frequentist operating characteristics of Bayesian adaptive designs are important in the context of clinical trials [34, 35]. For classical group sequential designs, several software packages are available that facilitate this [36]. For the Bayesian group sequential designs described here, the R package *gsbDesign* [9] allows to evaluate operating characteristics, such as the expected sample size and the probability of stopping at each of the interim analyses. For these calculations, we used normal approximations for various endpoints with known standard deviation. Although these simplifying assumptions are not necessary, they have the advantage of greatly reducing the computing time, especially for endpoints that would require the use of Markov chain Monte Carlo methods for the Bayesian analysis. For large clinical trials, the approximations are usually appropriate, and very similar approaches are also used for evaluating classical group sequential designs [5]. Although the normal approximations will typically be less appropriate for small clinical trials, the requirements on the precision of the operating characteristics are lower in earlier phases of drug development. It should also be noted that the impact of nuisance parameters (e.g., the assumed value of the standard deviation) on the operating characteristics is usually

far stronger than the effects because of the simplifying normal approximations.

The decision rules studied in this paper and implemented in gsbDesign are all formulated in terms of posterior probabilities. Alternatively, one could also work with rules that make predictive statements about the results at the next stages (see the discussion in [7], ch. 6.6.3). For example, the predictive power is often relevant when deciding on whether to stop a clinical trial for futility [37,38]. Although we focused here on posterior statements rather than on predictive ones, the framework applies to either of the statements.

An alternative approach for the design of group sequential designs is based on decision theory. Bayesian decision-theoretic designs can provide trial designs that are directly related to utility [39–41]. However, the specification of the utility is also more demanding, as various aspects of the decision context have to be formally quantified.

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